

DESIGN AN ELEVATOR

Overview

Req. gathering

Class diagram

Schema diagram

Code → H.W

- * Watch 1st 2 videos. playlist link in description
- * Notes and other resources are in Github.

Overview

- Align with interviewee
- Try to understand what interviewee is asking.

Design an elevator

you know the system

By elevator, do you mean the one commonly found in malls and buildings that is used to move people from one floor to another?

you don't know the system

Thank you for your question. Sorry, I am not familiar with the elevator system. Could you please provide a brief overview of it; that will help me understand the system and think about the requirements.

① Do you want me to

Requirement gathering

⑥ Design entities OR

⑦ Complete software system.

② Where can we store the data?

⑥ In-memory OR

⑦ In real database.

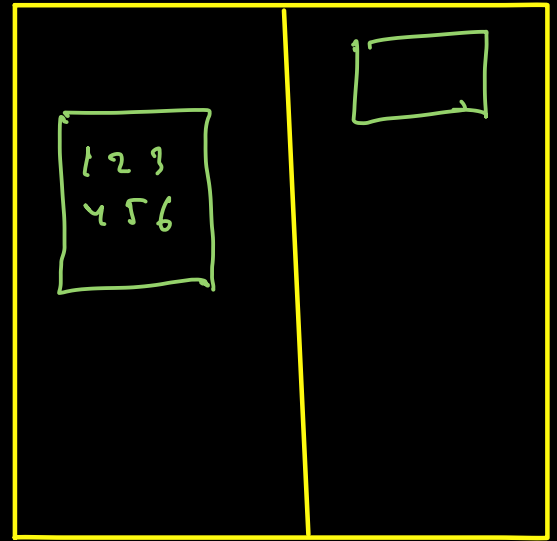
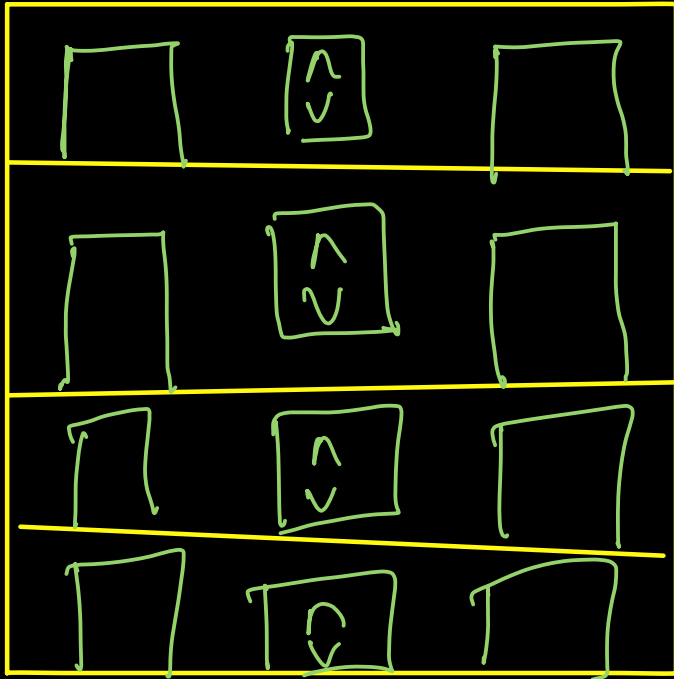
③ How user can interact with the system?

⑥ API OR

⑦ CMD OR

⑧ Hand code

REQUIREMENT GATHERING & DOUBT CLEARING



- ① A building can have multiple floors and elevators.
- ② User on any floor can press Up/Down
- ③ Passengers inside elevator can select one or more destination floor.
- ④ Elevator move and down based on request and current direction.

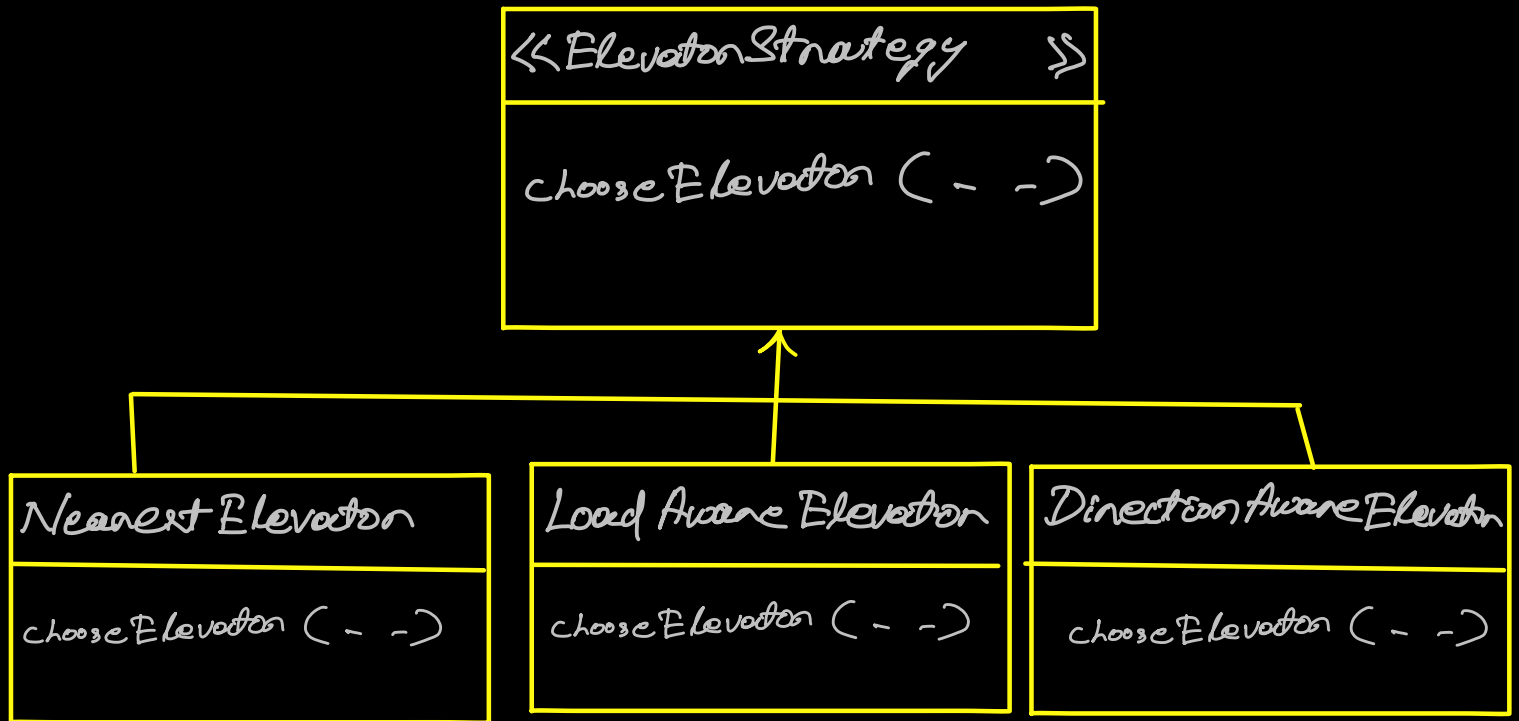
- ⑤ When multiple elevator exist, choose the best elevator for the request.
- ⑥ Open/Close elevator door at stops with safety check.
- ⑦ Prevent over load beyond a certain weight.
- ⑧ Elevator monitor should update with floor number, Move up, move down.

CLASS DIAGRAM

- ① Nouns
- ② Outer → Inner

Design Patterns

① Strategy Design Pattern



② Observer Design Pattern

